

Algorithms HW

1. Suppose that $f : \mathbb{C} \rightarrow \hat{\mathbb{C}}$ is a meromorphic function with two periods ω_1, ω_2 , such that the ratio ω_1/ω_2 is real.
 - (a) Suppose ω_1/ω_2 is rational. Show that the periodicity assumption is equivalent to being periodic with respect to some single ω' .
 - (b) If ω_1/ω_2 is irrational, show f is constant.

1. Let us write $\omega_1/\omega_2 = p/q$ for some coprime integers p, q . Let us define $\omega_0 := \omega_1/p$. We need to show that (1) f is periodic with respect to ω_0 and (2) that ω_1, ω_2 are multiples of ω_0 .

We show (2) first. Notice that $\omega_1 = p\omega_0$ by definition. And also, $\omega_2 = \frac{q\omega_1}{p} = q\omega_0$.

Next we show (1), that f is periodic with respect to ω_0 . Recall Bézout's identity gives us that there exists integers m, n such that $mp + nq = 1$ since p, q are coprime. Rearranging gives us that

$$\begin{aligned} \frac{\omega_1}{p} &= \omega_1 \left(m + n \frac{q}{p} \right) \\ &= \omega_1 \left(m + n \frac{\omega_2}{\omega_1} \right) \\ \omega_0 &= m\omega_1 + n\omega_2. \end{aligned}$$

Since f is periodic in ω_1 and ω_2 it thus follows that f is periodic in ω_0 .

2. Let $\tau := \omega_1/\omega_2$, we have that τ is irrational. Let $m, n \in \mathbb{Z}$ and notice that $m\omega_1 + n\omega_2 = \omega_2(m\tau + n)$. Now since τ is irrational we have that the set $\{\omega_2(m\tau + n)\}$ is dense in the line $\mathbb{R}\omega_2$. We will show that f is thus constant on the line $\mathbb{R}\omega_2$.

Let $c \in \mathbb{R}\omega_2$ be a point where f is holomorphic. Since $\{\omega_2(m\tau + n)\}$ is dense in $\mathbb{R}\omega_2$ we have a sequence $z_k = \omega_2(m_k\tau + n_k)$ with $m_k, n_k \in \mathbb{Z}$ which converges to c . Since f is holomorphic at c we have that it is in particular continuous at c , and so $f(c) = \lim_{k \rightarrow \infty} f(z_k)$. Moreover since $\omega_2(m_k\tau + n_k) = m_k\omega_1 + n_k\omega_2$ for all k we have that $f(\omega_2(m\tau + n)) = f(0)$ for all $m, n \in \mathbb{Z}$. But then $f(c) = \lim_{k \rightarrow \infty} f(z_k) = f(0)$, i.e. f is constant along $\mathbb{R}\omega_2$.

We have shown that f is constant on a closed subset of its domain of holomorphicity. It follows by the identity theorem that f is thus constant on its entire domain.

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3. Given a lattice $\Lambda \subset \mathbb{C}$, show that the series

$$\sum_{\omega \in \Lambda^*} \frac{1}{|\omega|^2}$$

is divergent.

Recall that without loss of generality we can take our lattice Λ to be generated by 1 and $\tau = a + bi$ with $a, b \in \mathbb{R}$ and $b > 0$. We can also choose generators of our lattice so that $a > 0$. With this notation we have that $\omega = m + n\tau$ for integers m, n , and so

$$|\omega|^2 = |m + n(a + bi)|^2 = (m + na)^2 + (nb)^2.$$

Our series of interest becomes

$$\sum_{\omega \in \Lambda^*} \frac{1}{|\omega|^2} = \sum_{m, n \in \mathbb{Z}} \frac{1}{(m + na)^2 + (nb)^2}.$$

For m, n not both simultaneously zero. Consider this sum for some fixed n . We can split our sum

$$\begin{aligned} \sum_{m, n \in \mathbb{Z}} \frac{1}{(m + na)^2 + (nb)^2} &= \sum_{(m+na)^2 \leq (nb)^2} \frac{1}{(m + na)^2 + (nb)^2} + \sum_{(m+na)^2 > (nb)^2} \frac{1}{(m + na)^2 + (nb)^2} \\ &\geq \sum_{(m+na)^2 \leq (nb)^2} \frac{1}{(m + na)^2 + (nb)^2} \\ &\geq \sum_{(m+na)^2 \leq (nb)^2} \frac{1}{2(nb)^2} \\ &= K(n) \cdot \frac{1}{2(nb)^2}. \end{aligned}$$

Where $K(n)$ is the number of integers m satisfying $(m + na)^2 \leq (nb)^2$. The second line follows since the latter sum on the first line contains only positive terms. The third line follows since, by the condition in the sum, $(nb)^2 \geq (m + na)^2$ for each term. Counting the number of integers m satisfying $(m + na)^2 \leq (nb)^2$ gives us that $K(n) \sim 2|nb|$ and so we have

$$\sum_{m, n \in \mathbb{Z}} \frac{1}{(m + na)^2 + (nb)^2} \geq \frac{1}{|nb|}.$$

Now we can show that our series of interest diverges

$$\sum_{\omega \in \lambda} \frac{1}{|\omega|^2} = \sum_{m,n \in \mathbb{Z}} \frac{1}{(m+na)^2 + (nb)^2} \geq \sum_{n \in \mathbb{Z}} \frac{1}{|nb|} = \frac{1}{b} \sum_{n \in \mathbb{Z}} \frac{1}{|n|},$$

which diverges.

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4

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